

32.7 Summary and Notes

32.7.1 Summary

Idea.

1. A factor is random when its levels are a sample and the conclusion is meant for the population of levels. The one-way random effects model has covariance $\sigma^2\mathbf{I} + \sigma_a^2\mathbf{Z}\mathbf{Z}^\top$, intraclass correlation $\sigma_a^2/(\sigma_a^2 + \sigma^2)$, and unbiased moment estimates from the analysis of variance (Definition 32.1.1 and Proposition 32.1.1).
2. In a balanced normal layout the between and within sums of squares are independent scaled chi-squares, so MSB/MSE has an exact F distribution for every value of the variance ratio. Inverting it gives an exact interval for σ_a^2/σ^2 and for the intraclass correlation (Corollary 32.1.2).
3. The moment estimate of a variance can be negative, with probability $\Pr\{F(g-1, n-g) < 1/(1+m\gamma)\}$, which is just over one half when $\sigma_a^2 = 0$. Reporting, truncating and re-thinking the model are all defensible responses (Eq. 32.1.4).
4. The linear mixed model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}$ has marginal mean $\mathbf{X}\boldsymbol{\beta}$ and covariance $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}$, a structured member of the general Gauss–Markov family. Woodbury’s identity inverts $\mathbf{V}$ at the cost of a $q \times q$ inverse (Definition 32.2.1 and Proposition 32.2.1).
5. Ordinary least squares is efficient exactly when $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, which holds for cluster-constant regressors in a balanced design and fails as soon as a regressor varies within clusters; the mixed estimate then interpolates between the within and between estimators (Example (Cluster-constant regressors) and Example (Regressors that vary within a cluster)).
6. The best linear unbiased predictor of $\boldsymbol{\lambda}^\top \boldsymbol{\beta} + \mathbf{c}^\top \mathbf{u}$ is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{c}^\top \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$: the best linear predictor with $\boldsymbol{\beta}$ replaced by its generalized least squares estimate. Its prediction error variance has two terms, one for predicting $\mathbf{u}$ and one for estimating $\boldsymbol{\beta}$ (Theorem 32.3.1, which is Theorem 14.2.8 of Section 14.2 written for mixed functions).
7. In the balanced one-way model the BLUP is $B(\bar{y}_k - \bar{y})$ with $B = m\gamma/(1+m\gamma)$: shrinkage towards the overall mean, of the empirical Bayes kind, with a genuine gain in mean squared error and a conditional bias that matters for ranking (Eq. 32.3.3).
8. Henderson’s mixed model equations solve for $(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}})$ in one sparse system of size $p+q$; they are the normal equations of a penalized least squares criterion with penalty $\mathbf{u}^\top \mathbf{G}^{-1}\mathbf{u}$, and the inverse of their coefficient matrix holds both $\text{Cov}(\hat{\boldsymbol{\beta}})$ and the prediction error variance (Theorem 32.3.2).
9. Restricted maximum likelihood is the likelihood of a maximal set of error contrasts. It does not depend on which contrasts are used, it equals the profile likelihood plus $-\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X})$ up to constants, and it reduces to $s^2 = \text{SSE}/(n-p)$ when $\mathbf{V} = \sigma^2\mathbf{I}$ (Definition 32.4.1 and Theorem 32.4.2).
10. In the balanced one-way model, the interior maximum likelihood estimate underestimates σ_a^2 by exactly $\lambda_1/(gm)$, while REML reproduces the unbiased moment estimates. The two score equations differ only in $\text{tr}(\mathbf{V}^{-1}\mathbf{V}_j)$ against $\text{tr}(\mathbf{P}\mathbf{V}_j)$ (Proposition 32.4.3 and Proposition 32.4.4).
11. Feasible generalized least squares is still unbiased when the variance estimate is translation invariant and even. Its Wald statistic is not normal: in a balanced cluster-randomized trial it is exactly $t(g-2)$, and treating it as normal costs real coverage.

Satterthwaite and Kenward–Roger approximate the right reference distribution in general (Proposition 32.5.1).

12. Testing $\sigma_a^2 = 0$ puts the null on the boundary. In the balanced one-way model the restricted likelihood ratio is an explicit increasing function of the F ratio, zero with probability $\Pr(F \leq 1)$, and its exact null distribution follows; asymptotically it is the chi-bar-squared mixture $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$ (Theorem 32.5.2, the variance-component companion of Proposition 32.5.1).
13. Restricted likelihoods may be compared only across covariance structures with the same fixed effects; comparisons of fixed effects need the ordinary likelihood (Section 32.5).
14. In the hierarchical reading, the BLUP is a posterior mean and Henderson’s coefficient matrix is a posterior precision; the restricted likelihood is the marginal posterior of the variance components under a flat prior on $\boldsymbol{\beta}$. A plug-in analysis omits the variance of the conditional mean, so its intervals are too short (Proposition 32.6.1).

32.7.2 Notes and sources

Fixed and random effects. The separation of variability into components goes back to Fisher (1918), who decomposed the correlation between relatives. Eisenhart (1947) fixed the terminology “model I” and “model II” and made explicit that the choice changes the expected mean squares and therefore which ratio is a test. Section 32.1 follows the expected-mean-squares route of Chapter 9, and Eq. 32.1.1 is the first of Henderson’s (1953) three methods for unbalanced data. Searle, Casella and McCulloch (1992) is the standard reference for variance component estimation, including methods this chapter omits such as Rao’s (1971) MINQUE; Burdick and Graybill (1992) treats intervals for variance components at book length.

Prediction. Theorem 32.3.1 is Section 14.2’s theorem with a random effect in place of the new observation’s error, which is why its proof is a substitution. Best linear unbiased prediction is due to Henderson, in a sequence of papers beginning with the abstract Henderson (1950) and developed in Henderson, Kempthorne, Searle and von Krosigk (1959) and Henderson (1975); the motivating problem was ranking dairy sires from unbalanced records, where shrinkage is the difference between a usable and an unusable ranking. Harville (1976, 1977) gave the modern treatment, including the relation to Bayesian prediction. Robinson (1991) is the best single essay on what BLUP is and why several derivations converge on it, and the warning in Section 32.3 about ranking shrunk predictions is developed there. The equivalence with penalized least squares, and hence with ridge regression (Theorem 27.2.2), is the bridge to smoothing: Ruppert, Wand and Carroll (2003) build semiparametric regression on it, and Chapter 43 uses it to estimate a smoothing parameter by REML (Theorem 43.6.1).

REML. Restricted maximum likelihood was introduced by Patterson and Thompson (1971) for recovering inter-block information in incomplete block designs, the problem Yates (1940) had posed; Section 18.5 met it in its fixed-effects form. Harville (1974) proved the marginal-posterior characterization that Proposition 32.6.1(b) restates, and Harville (1977) surveyed the area; Hartley and Rao (1967) had treated maximum likelihood for the mixed model a few years earlier, and Corbeil and Searle (1976) established REML’s advantage for small designs by simulation. The determinant identity Eq. 32.4.5 and the identity $\mathbf{K}(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top = \mathbf{P}$ are standard; the proof of the latter given here, through a constrained minimization, seems the shortest route and makes Lemma 32.4.1(c) immediate. On algorithms, Dempster, Laird and Rubin (1977) supplied EM, Laird and Ware (1982) brought it to longitudinal data, and Gilmour, Thompson and Cullis (1995) introduced average information REML. Bates, Mächler,

Bolker and Walker (2015) describe a sparse-matrix implementation in which the profiled deviance is evaluated by a single sparse Cholesky update.

Inference. The unbiasedness result Proposition 32.5.1(a) is the covariance analogue of the weighted least squares argument of Section 21.3; Kackar and Harville (1981) proved it in this generality, and Kackar and Harville (1984) gave the approximation to the extra variance of $\hat{\beta}(\hat{\theta})$ that Kenward and Roger (1997) turned into a usable small-sample correction. The degrees-of-freedom approximation of Eq. 32.5.2 is Satterthwaite (1946); Giesbrecht and Burns (1985) extended it to mixed models. For the boundary problem, Self and Liang (1987) gave the general asymptotic theory of likelihood ratios when the true parameter is on the boundary, and Stram and Lee (1994) worked out the mixed-model cases most often needed. Crainiceanu and Ruppert (2004) showed that for a single variance component the exact finite-sample null distribution of the restricted likelihood ratio can be computed, and documented how poor the chi-bar-squared approximation can be; the exact balanced calculation of Theorem 32.5.2 is a special case that can be done by hand, and the proof sketch of the chi-bar-squared limit given after it is our own. One question left to the econometric literature is whether the random effects are uncorrelated with the regressors at all, as Definition 32.2.1 assumes; if they are not, the mixed fit is inconsistent and the within fit of Example (Regressors that vary within a cluster) is not. Hausman (1978) tests this by comparing the two, using the fact that when one estimator is the BLUE the covariance of the difference is the difference of the covariances. Verbeke and Molenberghs (2000) and Pinheiro and Bates (2000) are the standard applied references, the first for the theory of inference in linear mixed models and the second for practice; McCulloch, Searle and Neuhaus (2008) covers both linear and generalized mixed models.

Bayesian mixed models. Gelman (2006) changed practice on priors for variance parameters, showing that the inverse gamma with small parameters is not a safe default and recommending half- t priors on standard deviations; the scale-mixture representation used in our sampler is from that paper.

Sources for the chapter. The coverage checklist was built from Rencher and Schaalje (2008, chapter 17), Agresti (2015, sections 9.2–9.3), Fahrmeir, Kneib, Lang and Marx (2021, sections 7.1–7.4) and Sen and Srivastava (1990, sections 7.4–7.5). All the text, proofs, examples, data and exercises here are our own. The proficiency study is simulated from stated parameter values with a fixed seed; the investment panel is Grunfeld’s, in the public-domain reconstruction distributed with statsmodels, and continues the analysis begun in Section 6.6.

32.7.3 References

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